METRIC

DOD-P-24631/1B (SH)

3 February 1992
SUPERSEDING
DOD-P-24631/1A(SH)
16 October 1987

MILITARY SPECIFICATION SHEET

PAINT, EPOXY, BLACK, NAVY FORMULA 184, TYPE 1 (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Department of Defense.

The requirements for acquiring the paint described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standard (DODISS) specified in the solicitation: DOD-P-24631.

Formula: This formula covers black epoxy paint, designated Navy Formula 184. The paint shall consist of the ingredients specified in the quantities specified.

AMSC N/A FSC 8010

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TABLE I. Navy formula 184 - component A.

			Pounds	Gallons
* Curing agent	2216A (Transparent) 3M Company	1/	218.8	21.18
Iron Black	Irox Black 724, Reichard-Coulston	2/	39.5	0.95
Amorphous Silica	Imsil A-15, Illinois Mineral	3/	98.7	4.48
Diatomaceous Silica	Celite White Mist, Johns-Manville	4/	65.8	3.43
Dispersant	Tenlo 70, Henkel	5/	0.7	0.09
Thickener	Bentone SD-2, Rheox, Inc.	6/	5.3	0.39
Naphtha	Aromatic 100 Union Chemical	7/	21.1	2.90
n-Butanol	n-Butyl alcohol, Chemcentral	8/	111.9	16.58
			561.8	50.00

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TABLE II. Navy formula 184 - component B.

				Pounds	Gallons
*	Epoxy Resin	2216B (Clear Amber) 3M	9/	185.6	20.24
	Iron Black	Irox Black 724, Reichard-Coulston	2/	39.5	0.95
	Amorphous Silica	Imsil A-15, Illinois Mineral	3/	98.7	4.48
	Diatomaceous Silica	Celite White Mist, Johns-Manville	4/	46.1	2.40
	Dispersant	Tenlo 70, Henkel	5/	0.7	0.09
	Thickener	Bentone SD-2, Rheox, Inc.	6/	5.3	0.39
	Naphtha	Aromatic 100 Union Chemical	7/	19.7	2.72
	n-Butanol	n-Butyl alcohol, Chemcentral	8/	126.4	18.73
				522.0	50.00

NOTES:

* 1. The curing agent shall be Scotch Weld 2216A (transparent), available from the 3M Company, 3M Center, 2501 Hudson Road, St. Paul, MN 55144. The infrared spectra of the curing agent shall match the infrared spectra given in Figure 1. There is no substitute for this curing agent which provides the flexible property to this paint.

* 2. The iron black shall be Irox Black 724, available from Reichard-Coulston, Inc., 1421 Mauch Chunk Road, Bethlehem, PA 18018, or equivalent.

* 3. The amorphous silica shall be Imsil A-15, available from the Illinois Minerals Company, 2035 Washington Avenue, Cairo, IL 6914, or equivalent.

* 4. The diatomaceous silica shall be Celite White Mist, available from the Johns-Manville Corporation, P.O. Box 5108, Denver, CO 80217, or equivalent.

* 5. The dispersant shall be Tenlo 70, available from the Henkel Corporation, 300 Brookside Avenue, Ambler, PA 19002, or equivalent.

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NOTES: (Continued)

* 6. The thickener shall be Bentone SD-2, available from Rheox, Inc., P.O. Box 700, Hightstown, NJ 08520, or equivalent.

* 7. The aromatic naphtha shall be any naphtha with an aromatic content of 97-99% and a minimum flash point of 110° F (43° C) when measured in accordance with ASTM D-3278.

* 8. Normal-butyl alcohol (n-butyl alcohol) shall have a specific gravity or 0.807 to 0.811 and a flash point greater than 95° F (35° C) when measured in accordance with ASTM D-3278.

9. The epoxy resin shall be Scotch Weld 2216B (clear amber), available from the 3M Company. 3M Center, 2501 Hudson Road, St. Paul, MN 55144, or equivalent such as Epon 828 from Shell Chemical Co. The infrared spectra of the epoxy resin shall match the infrared spectra given in Figure 2.

Use of alternate ingredients in the formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that the paint compound made using the alternative ingredient will conform to all requirements of this specification.

QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of Section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed components tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed Components	
	Min	Max	Min	Max	Min	Max
Pigment content, percent by weight	36.3	38.3	35.3	37.3	---	---
Pigment content, percent by volume	17.5	19.5	15.5	17.5	---	---
Volatiles, percent by weight	22.8	23.7	27.1	27.9	---	---
Volatiles, percent by volume	37.9	39.0	41.8	49.9	---	---
Nonvolatile vehicle, percent by weight	37.9	39.9	34.6	36.6	---	---
Nonvolatile vehicle, percent by volume	41.4	43.4	39.5	41.5	---	---

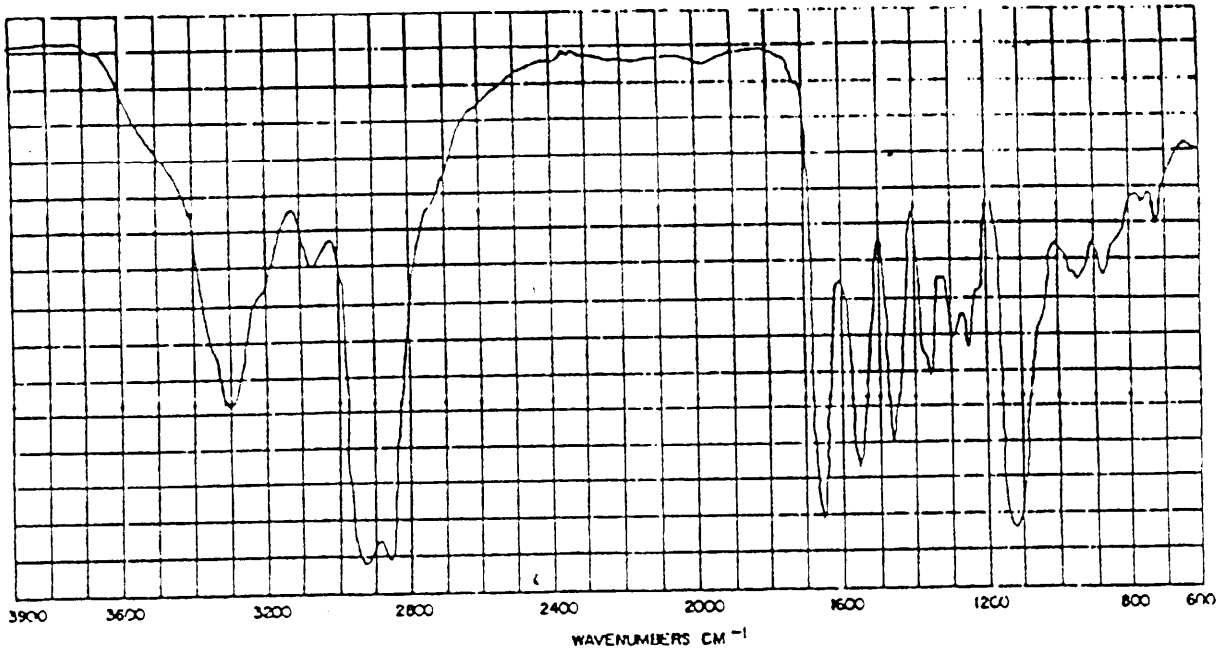
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QUANTITATIVE REQUIREMENTS. Continued.

Consistency, KU	---	---	---	---	66	---	
Pounds per gallon	10.9	11.5	10.1	10.7	10.5	11.1	
Set-to-touch time, hours at 73° F	---	---	---	---	---	6.0	
Volatile Organic Content (VOC) G/L (Lbs/Gal)	---	---	---	---	---	340	(2.8)
Dry hard, hours at 73° F	---	---	---	---	---	24.0	
Fineness of grind, Hegman	---	---	---	---	6.0	---	
Flash Point, ASTM D-3278	---	---	---	---	80	---	
Pot life, hours at 73° F	---	---	---	---	3	---	
Gloss, 20° specular, percent	---	---	---	---	---	1	
Gloss, 60° specular, percent	---	---	---	---	---	4	
Gloss, 85° specular, percent	---	---	---	---	---	5	
Wet Reflectance (R_3), percent	---	---	---	---	0.80	1.02	
Contrast ratio, 3 mils dry film	---	---	---	---	0.91	---	

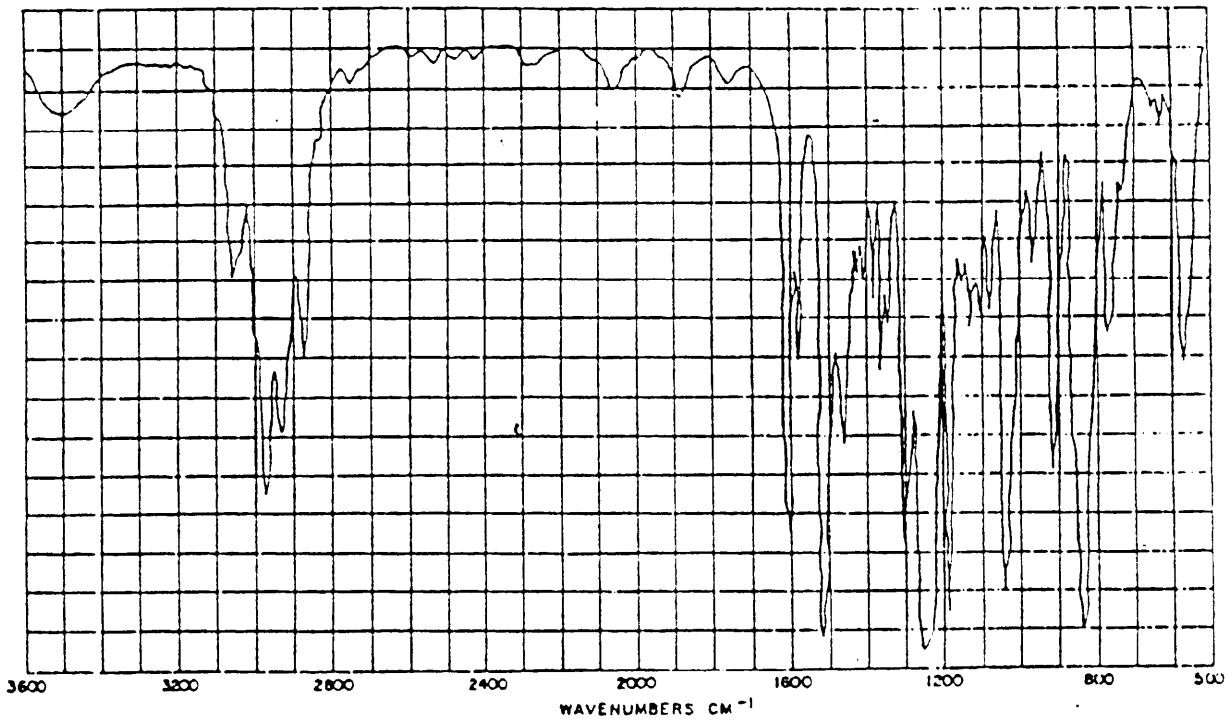
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SH 12139

Figure 1. Infrared spectrum of Scotch Weld 2216A aliphatic amine.



SH 12140

Figure 2. Infrared spectrum of Scotch Weld 2216B epoxy resin.

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